1

October 27, 2025

1 ” ” (-)

1.1 1.

: Anaconda
: Jupyter Notebook
: Python 3.7
: Pandas ( ), NumPy ( ), Matplotlib Seaborn ( )

2 2.

2.1 2.1.1

Pandas read_csv () iris.csv df DataFrame df.info() info () iris 150 5 sepal_length sepal_width petal_length petal_width 4 float64 species object “Non-Null Count” 150

[13]:
```python
import seaborn as sns
import pandas as pd
import numpy as np
df_iris = sns.load_dataset('iris')
df_iris.to_csv('iris.csv', index=False)
print("'iris.csv'       E:\\    ")
df = pd.read_csv("iris.csv")
print(df.shape)
print(df.head())
print("     (    )")
df.info()
print("\n   ( ,  )")
print(df.shape)
```

```
'iris.csv'       E:\
(150, 5)
   sepal_length  sepal_width  petal_length  petal_width species
0           5.1          3.5           1.4          0.2  setosa
1           4.9          3.0           1.4          0.2  setosa
2           4.7          3.2           1.3          0.2  setosa
3           4.6          3.1           1.5          0.2  setosa
4           5.0          3.6           1.4          0.2  setosa
```

```
    (    )
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 150 entries, 0 to 149
Data columns (total 5 columns):
 #   Column        Non-Null Count  Dtype
---  ------        --------------  -----
 0   sepal_length  150 non-null    float64
 1   sepal_width   150 non-null    float64
 2   petal_length  150 non-null    float64
 3   petal_width   150 non-null    float64
 4   species       150 non-null    object
dtypes: float64(4), object(1)
memory usage: 6.0+ KB

   ( ,  )
(150, 5)
```

2.2 2.1.2

species .value_counts() 'setosa' 'versicolor' 'virginica' 50

[14]:
```python
print("\n     ")
species_counts = df['species'].value_counts()
print(species_counts)
```

```
setosa        50
versicolor    50
virginica     50
Name: species, dtype: int64
```

2.3 2.2.1 ( )

numpy np.nan .loc 10 11 12 13 14 5 petal_width NaN

df.info() petal_width "Non-Null Count" 145 150 5

[15]:
```python
import numpy as np
df.loc[10:14, 'petal_width'] = np.nan
print("      ")
df.info()
print("\n       ")
print(df.iloc[9:16])
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 150 entries, 0 to 149
```

```
Data columns (total 5 columns):
 #   Column        Non-Null Count  Dtype  
---  ------        --------------  -----  
 0   sepal_length  150 non-null    float64
 1   sepal_width   150 non-null    float64
 2   petal_length  150 non-null    float64
 3   petal_width   145 non-null    float64
 4   species       150 non-null    object 
dtypes: float64(4), object(1)
memory usage: 6.0+ KB
```

```
    sepal_length  sepal_width  petal_length  petal_width species
9            4.9          3.1           1.5          0.1  setosa
10           5.4          3.7           1.5          NaN  setosa
11           4.8          3.4           1.6          NaN  setosa
12           4.8          3.0           1.4          NaN  setosa
13           4.3          3.0           1.1          NaN  setosa
14           5.8          4.0           1.2          NaN  setosa
15           5.7          4.4           1.5          0.4  setosa
```

2.4 2.2.2 “ ”

petal_width NaN median_value 1.3 df ['petal_width'].fillna (median_value, inplace=True) inplace=True df df.info() petal_width “Non-Null Count” 150

[16]:
```python
mean_value = df['petal_width'].mean()
print(f"    : {mean_value}")
median_value = df['petal_width'].median()
print(f"     : {median_value}")
df['petal_width'].fillna(median_value, inplace=True)
print("\n     ")
df.info()
print("\n    9   15  ")
print(df.iloc[9:16])
```

```
    : 1.2351724137931035
     : 1.3


<class 'pandas.core.frame.DataFrame'>
RangeIndex: 150 entries, 0 to 149
Data columns (total 5 columns):
 #   Column        Non-Null Count  Dtype  
---  ------        --------------  -----  
 0   sepal_length  150 non-null    float64
```

```
 1   sepal_width   150 non-null    float64
 2   petal_length  150 non-null    float64
 3   petal_width   150 non-null    float64
 4   species       150 non-null    object
dtypes: float64(4), object(1)
memory usage: 6.0+ KB

    9   15
    sepal_length  sepal_width  petal_length  petal_width species
9            4.9          3.1           1.5          0.1  setosa
10           5.4          3.7           1.5          1.3  setosa
11           4.8          3.4           1.6          1.3  setosa
12           4.8          3.0           1.4          1.3  setosa
13           4.3          3.0           1.1          1.3  setosa
14           5.8          4.0           1.2          1.3  setosa
15           5.7          4.4           1.5          0.4  setosa
```

2.5 3.3.1 & 3.3.2 “ ” “ ”

Pandas sepal_length sepal_width sepal_area petal_length pe
df ['sepal_area'] = df ['sepal_length'] * df ['sepal_width'] df ['petal_area'] = df ['petal_length'] * df ['petal_width']
df.head () 5 sepal_area petal_area

[17]:
```python
df['sepal_area'] = df['sepal_length'] * df['sepal_width']
df['petal_area'] = df['petal_length'] * df['petal_width']
print("      ")
print(df.head())
```

```
   sepal_length  sepal_width  petal_length  petal_width species  sepal_area  \
0           5.1          3.5           1.4          0.2  setosa       17.85
1           4.9          3.0           1.4          0.2  setosa       14.70
2           4.7          3.2           1.3          0.2  setosa       15.04
3           4.6          3.1           1.5          0.2  setosa       14.26
4           5.0          3.6           1.4          0.2  setosa       18.00

   petal_area
0        0.28
1        0.28
2        0.26
3        0.30
4        0.28
```

2.6 2.3.3

df.corr () 6
6x6 - 1 1 1 -1 0

[18]:
```python
correlation_matrix = df.corr()
print("---      ---")
print(correlation_matrix)
```

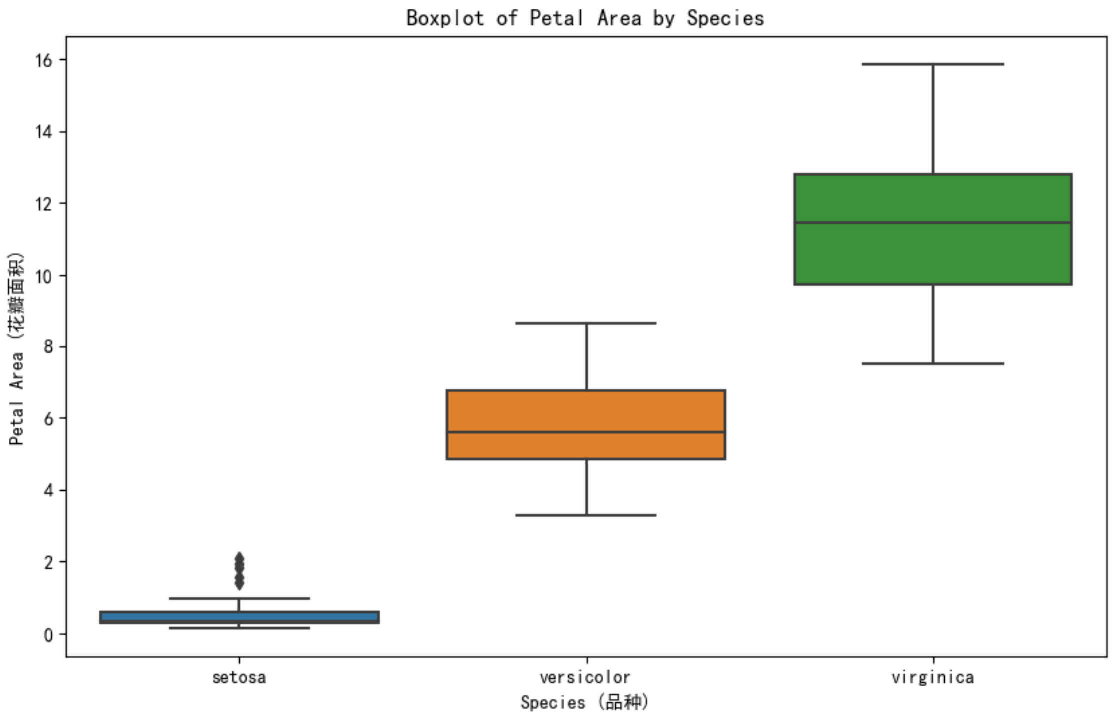
```
---      ---
              sepal_length  sepal_width  petal_length  petal_width  \
sepal_length      1.000000    -0.117570      0.871754     0.793019
sepal_width      -0.117570     1.000000     -0.428440    -0.337107
petal_length      0.871754    -0.428440      1.000000     0.924878
petal_width       0.793019    -0.337107      0.924878     1.000000
sepal_area        0.679180     0.643461      0.360909     0.372184
petal_area        0.855937    -0.282685      0.954383     0.963970

              sepal_area  petal_area
sepal_length    0.679180    0.855937
sepal_width     0.643461   -0.282685
petal_length    0.360909    0.954383
petal_width     0.372184    0.963970
sepal_area      1.000000    0.457881
petal_area      0.457881    1.000000
```

2.7 2.4.1 “ ” ( )

Seaborn sns.boxplot (x=‘species’, y=‘petal_area’, data=df) “ ” ‘setosa’ ‘versicolor’ ‘virginica’ “ ”

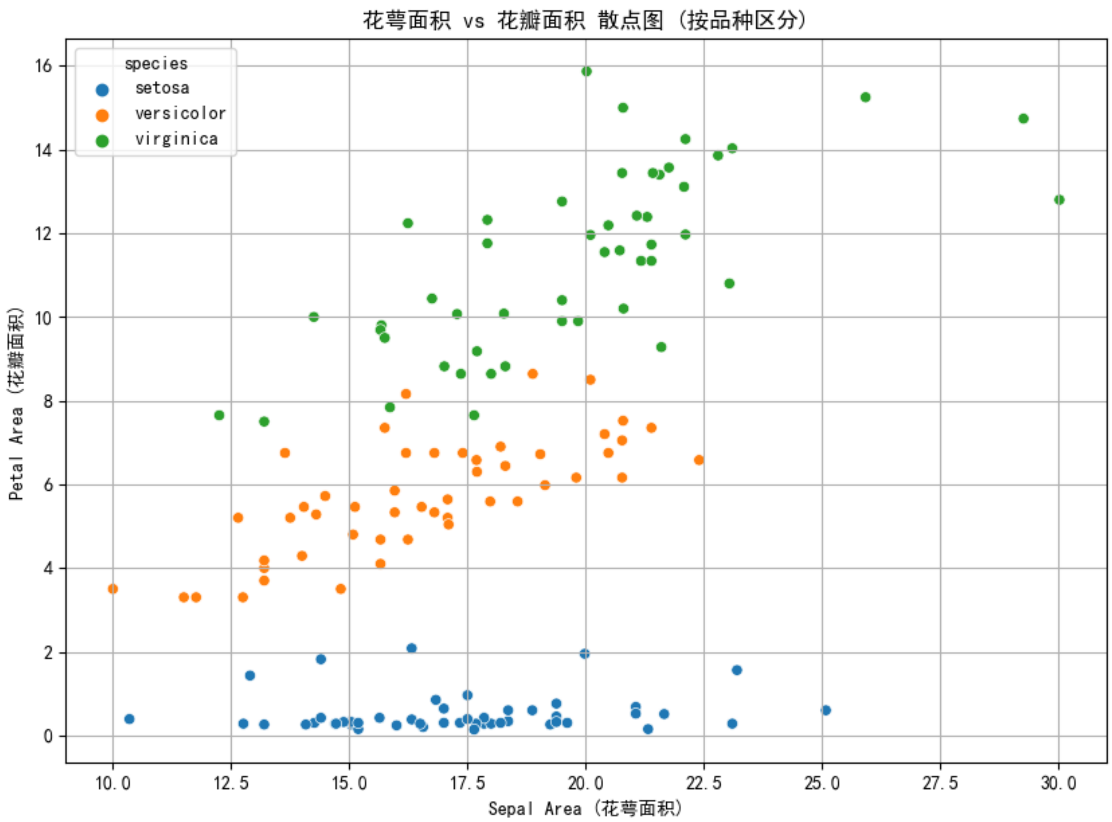
[19]:
```python
import matplotlib.pyplot as plt
import seaborn as sns
plt.rcParams['font.sans-serif'] = ['SimHei']
plt.rcParams['axes.unicode_minus'] = False
plt.figure(figsize=(10, 6))
sns.boxplot(x='species', y='petal_area', data=df)
plt.title('Boxplot of Petal Area by Species')
plt.xlabel('Species ( )')
plt.ylabel('Petal Area (   )')
plt.show()
```



2.8 2.4.2 " vs "

Seaborn sns.scatterplot () sepal_area x petal_area y hue='species' plt.grid (True) 'setosa' 'versicolor' 'virginica'

[20]:
```
plt.figure(figsize=(10, 7))
sns.scatterplot(x='sepal_area', y='petal_area', hue='species', data=df)
plt.grid(True)
plt.title('   vs      (   )')
plt.xlabel('Sepal Area (   )')
plt.ylabel('Petal Area (   )')
plt.show()
```



2.9 4.3.

Matplotlib plt.imshow () correlation_matrix plt.colorbar () plt.xticks plt.yticks 6 1.0 petal_length petal_area 1.0 sepal_width petal_length petal_width sepal_width

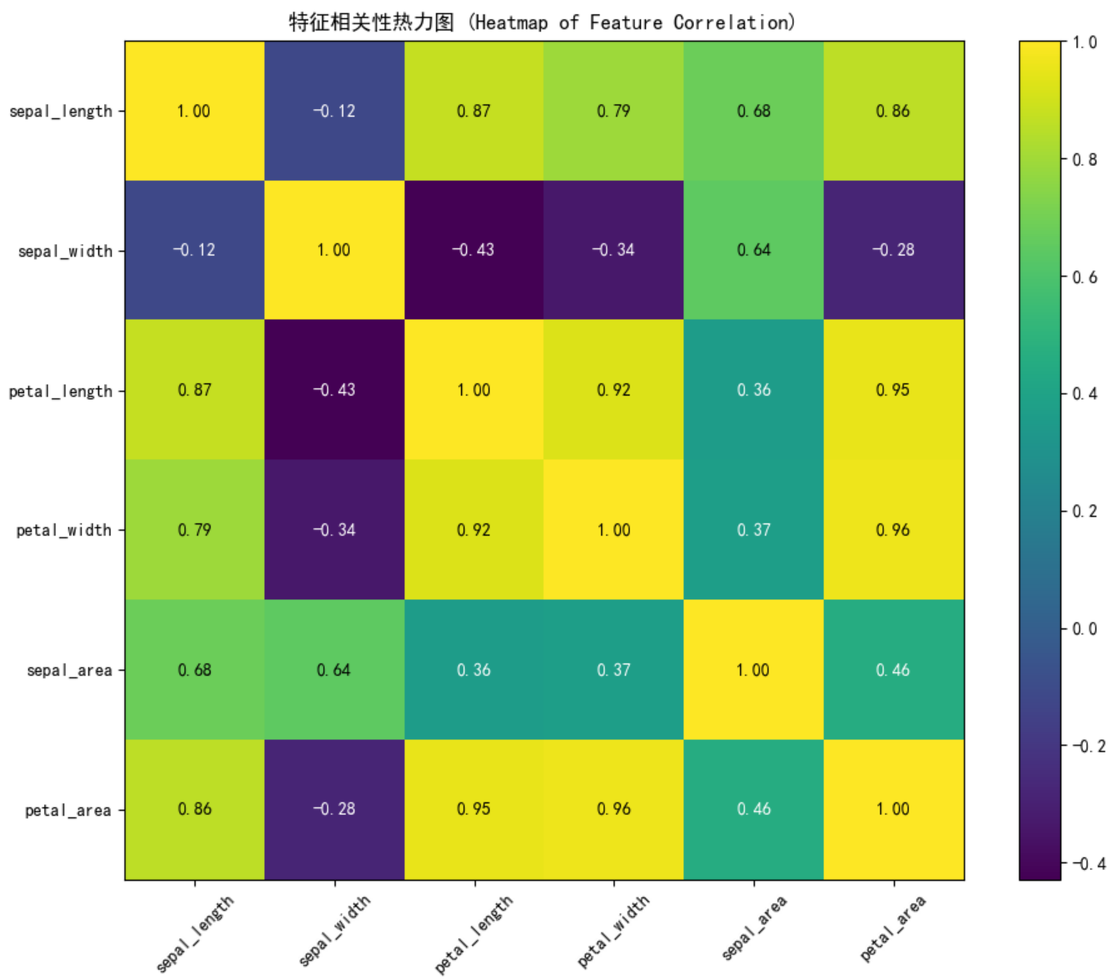
[21]:
```
plt.figure(figsize=(10, 8))
im = plt.imshow(correlation_matrix, cmap='viridis', interpolation='nearest')
plt.colorbar(im)

labels = correlation_matrix.columns
plt.xticks(ticks=range(len(labels)), labels=labels, rotation=45)
plt.yticks(ticks=range(len(labels)), labels=labels)

plt.title('      (Heatmap of Feature Correlation)')


for i in range(len(labels)):
```

```
    for j in range(len(labels)):
        text = plt.text(j, i, f'{correlation_matrix.loc[labels[i], labels[j]]:.2f}',
                        ha='center', va='center', color='w' if correlation_matrix.loc[labels[i], labels[j]] < 0.6 else 'black')
plt.tight_layout()
plt.show()
```



3

“ ”

[22]:
```
jupyter nbconvert --to pdf your_notebook.ipynb
```

```
  File "C:\Users\25405\AppData\Local\Temp\ipykernel_36164\1738622736.py", line
    jupyter nbconvert --to pdf your_notebook.ipynb
```

```
^
SyntaxError: invalid syntax
```

```
[ ]:
```